Alan Huang

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Education

Stanford University — September 2024 – May 2028
Bachelor of Science in Computer Science — *Stanford, CA*

- GPA: 3.96 / 4.00

Stanford University — September 2026 – May 2028
Master of Science in Computer Science (Coterminal) — *Stanford, CA*

Experience

Software Engineer Intern — June 2026 – August 2026
Walmart Global Tech — *Bentonville, AR*

- Built a Python internal tool for the Walmart Fulfillment Services (WFS) team, used by 40 seller-support agents to look up order status and shipping exceptions, cutting average issue resolution time from 15 to 5 minutes by consolidating queries across 6 backend systems
- Shipped 8 bug fixes and 15 test cases to a production Java-based order-tracking service handling 500,000 orders/day, merged via standard PR review with senior engineers
- Partnered with product managers across daily standups to translate seller feedback into 7 actionable feature requests for the fulfillment dashboard

AI Research Intern – NSF REU — June 2025 – August 2025
National Science Foundation (REU, hosted at Purdue University) — *Indianapolis, IN*

- Selected for the NSF REU Site: Enhancing Undergraduate Experiences in Data and Mobile Cloud Security, focusing on cybersecurity, mobile/cloud computing, and AI-driven data security
- Improved cross-dataset deepfake detection accuracy from 59.7% to 65.5% on the CelebDF benchmark by designing a diffusion-based Synthetic Domain Alignment technique, validated across FaceForensics++ and CelebDF
- Published as co-first author: *Robust Deepfake Detection via Synthetic Domain Alignment*, 22nd IEEE International Conference on Mobile Ad-Hoc and Smart Systems (IEEE MASS 2025)

Projects

OS: Crash-Consistent Filesystem & Encrypted Virtual Memory | *C, C++* — 2026

- Implemented a Unix V6 filesystem reader in C with write-ahead journaling and crash recovery, restoring consistency after simulated mid-write failures
- Built a memory-mapped encrypted file system in C++ using the Clock page-replacement algorithm to manage 128 physical memory pages, achieving a 27% reduction in page faults compared to a naive FIFO baseline
- Built for Stanford CS111: Operating Systems

Multi-Label Hate Crime Bias Classification | *Python, scikit-learn, XGBoost* — 2026

- Trained and benchmarked five classifiers (Logistic Regression, Decision Tree, Random Forest, XGBoost, majority baseline) to predict bias motivation category across 32,508 California hate crime incidents
- Addressed severe class imbalance (three rarest classes <3% of data) using SMOTE and stratified splitting, more than doubling macro F1 from 12.4% (majority-class baseline) to 25.0% with L2-regularized Logistic Regression
- Built for Stanford CS129: Applied Machine Learning — github

Publications

Alan Huang, Ian Yee, Shu Hu. *Robust Deepfake Detection via Synthetic Domain Alignment.* The 22nd IEEE International Conference on Mobile Ad-Hoc and Smart Systems (IEEE MASS 2025). DOI

Alan Huang, Shaheer Raza, Yuni Xia. *Predicting Diabetes with Medical and Demographic Data.* The 6th International Conference on Computer Science, Engineering, and Education (CSEE 2025). DOI

Technical Skills

Languages: Python, C/C++, Java, JavaScript, SQL
ML/Data: TensorFlow, scikit-learn, XGBoost, pandas, NumPy
Systems & Tools: Git, Node.js, FastAPI, Express, Supabase